Praktikum Design of Wind Farms

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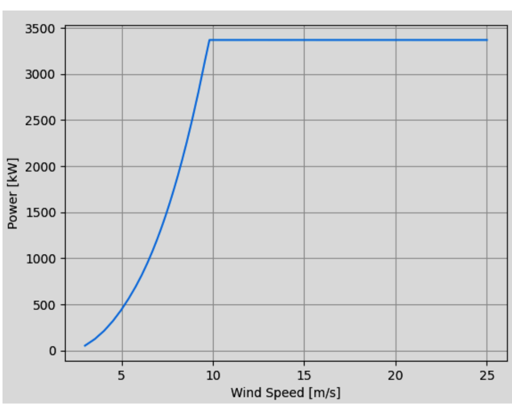
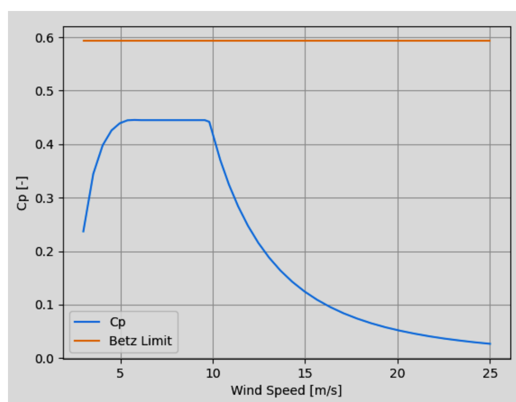
Assignment 2 (5%): Jensen's Wake Model and Wake Superposition in Wind Farms

This assignment aims at introducing students to the use of Jensen's wake model, the linear velocity deficit model, and the root-sum-square superposition model for the estimation of wind farm power production.

Three IEA 3.4MW wind turbines are planned to be installed in a row (north to south) with a separation distance of 5D between each successive turbine. The key parameters of the turbines and their characteristics are as follows:

Item	Value	Units
Name	IEA 3.4MW Reference	N/A
Rated Power	3370	kW
Rated Wind Speed	9.8	m/s
Cut-in Wind Speed	3	m/s
Cut-out Wind Speed	25	m/s
Rotor Diameter	130	m
Hub Height	120	m
Control	Pitch Regulated	N/A





A .csv file is available on GitHub.

1. Use Jensen's wake model to calculate by hand the relative and absolute power output of each wind turbine (P_n/P_1, for n=1, 2, and 3) when the wind speed is 7.5 m/s for two wind directions of 0° and 270°. Assume that k_w = 0.075 and the thrust coefficient is C_t = 0.766. Use the linear velocity deficit model, and the root-sum-square superposition model. Please upload the steps of your solution in Word or PDF.